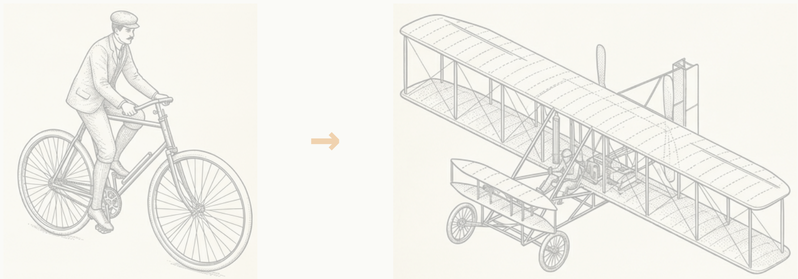
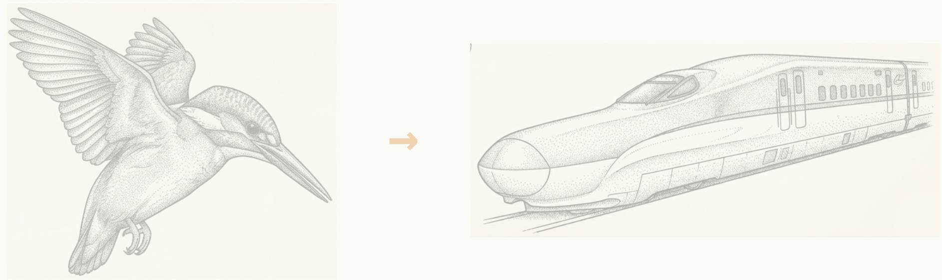
1903 FIG. 01



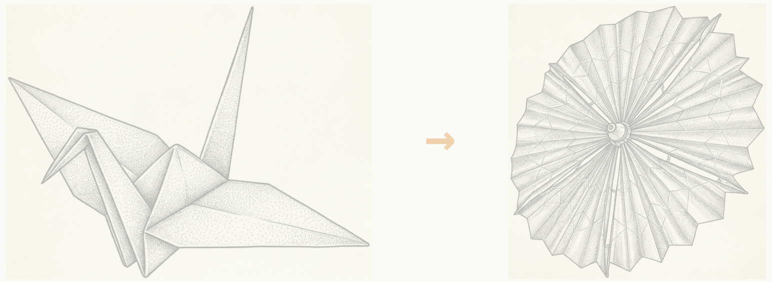
BICYCLE → FLYING MACHINE

1997 FIG. 02



KINGFISHER BEAK → SHINKANSEN NOSE

2016 FIG. 03



ORIGAMI → SOLAR SAIL

Carnegie Mellon University | Human-Computer Interaction Institute

KITTUR LAB · HUMAN-COMPUTER INTERACTION INSTITUTE

Solving industry’s hardest problems with solutions from *unexpected places*.

What the research offers, which companies to bring us, and how it can help you open doors.

Aniket Kittur · Professor, HCII

FOR CMU CORPORATE RELATIONS · SEPTEMBER 2026

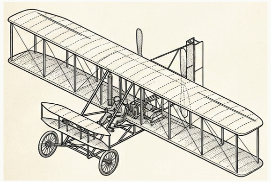
THE PROBLEM EVERY COMPANY BRINGS

Every R&D team is stuck on problems they're making only *incremental progress* on.

Because experts search where they already know — and AI gets you more of what's already out there.

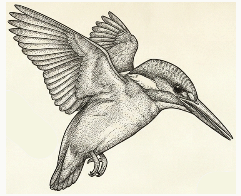
Our research makes finding breakthroughs from distant domains *systematic*: retrieve by mechanism, not keyword; map it into their domain; de-risk it into something their team can build and test.

BREAKTHROUGHS COME FROM ANALOGIES


→


Bicycle → Wright Flyer (1903)

Frame-lean steering became wing-warp steering: the control problem that grounded every prior aviator.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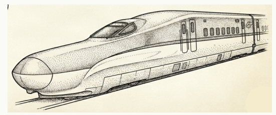
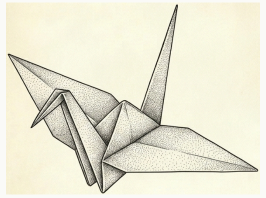


Kingfisher → bullet train (1997)

A beak that enters water without a splash became a nose that ends sonic booms in tunnels.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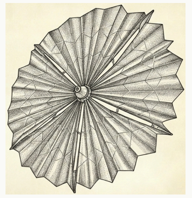


Origami → NASA solar sail (2016)

A Miura fold packs a 25 m sail into a 1 m fairing: a packaging problem became a paper-folding one.

Serendipity found these. We make them searchable — forward, and in reverse (a capability → the problems it could solve).

12+ years of research at HCII *PNAS* · best papers at CHI, CSCW, KDD

NSF-funded · 4 patents Kittur · Martelaro · Mohanty · Shahaf

WHAT IT LOOKS LIKE · 60 SECONDS

A public P&G challenge: *clean between teeth without floss.*

Nobody in oral care would look at an oil rig. The engine did – and came back with a concept, its evidence, and the tests to run.

The concept: food-safe iron-oxide microbeads (E172) self-assemble under a rotating field into spinning soft augers that scrub the gaps, then rinse away.

01 SEARCH

Retrieve by *mechanism*: "dislodge a stubborn deposit from a narrow channel." Matches came from **drilling fluids** that scour a borehole and **vascular medicine**, where magnetic bead swarms clear clots.

02 TRANSFER

Map the mechanism onto the real constraints: safe in the mouth, no hardware a consumer won't use, a sachet and a rinse. Every claim is tied to a primary source – not a hallucinated one.

03 EVALUATE

Hand back the four test protocols and success criteria that settle the risk: chain formation, biofilm softening, debris removal, bead recovery. The team knows what to run Monday.

The same loop runs on a materials scientist's holy-grail problem or a propulsion chemist's storage problem – with their experts in the loop.

PROOF · WITH INDUSTRY PARTNERS

Ideas their own experts rated as ones *they couldn't have reached*.

COVESTRO · FORTUNE 500 MATERIALS

3 "HOLY-GRAIL" PROBLEMS STUCK 10+ YEARS · 17 SCIENTISTS

54%

more novel, no drop in feasibility ($p < .05$)

30+

promising ideas vs. 0–1 from their own LLM tools

2

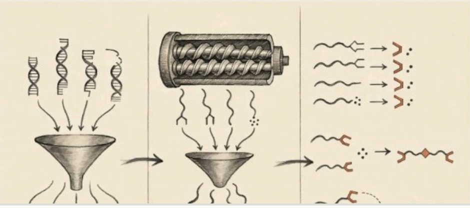
already in their managed innovation pipeline

IDEA 01 OF 03 NEXT ›

BORROWED FROM DNA-SEQUENCING PREPARATION

Convergent chain repair: boron funnels every broken end into one joinable handle

Borrow the terminus-funneling step from next-generation sequencing library preparation to converge every broken polyolefin chain end onto one boron handle and rebuild linear molecular weight without radicals or gel.



HOW IT WORKS

Thexylborane, carrying exactly two B–H bonds, hydroborates terminal vinyl and vinylidene chain ends in the melt, capturing two chains per molecule onto one dialkylborane handle; a solid-acid co-additive funnels carbonyl-derived ends into the same pathway via dehydration. Downstream CO injection drives Brown carbonylation [3], migrating both chains onto a carbonyl carbon to yield a linear ketone bridge – purely ionic, no radical generated, mid-chain attack impossible. No one has combined

"If we sat down and spent a week going through these things, that would be a worthwhile investment to make sure that we're spending the next two years on the right projects." — R&D group leader

URSA MAJOR · AEROSPACE PROPULSION "HOW CAN WE EXTEND THE LIFE OF OUR PROPELLANT?"

+80%

estimated propellant life at minimal extra cost vs. 5–10% from their previous approach

Paid pilot underway.

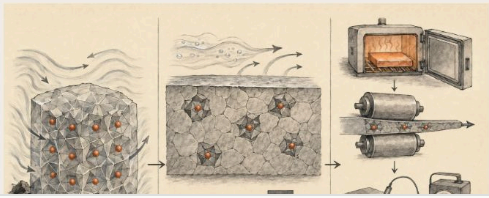
Concepts borrowed from semiconductor wafer gettering and nuclear-waste ceramics; scope expanding.

IDEA 03 OF 03 ‹ PREV

BORROWED FROM NUCLEAR-WASTE CERAMICS

Lattice-pinned alloy: heat treatment locks trace metals where peroxide cannot reach

Borrow lattice-site immobilization from nuclear-waste ceramics to trap catalytic trace metals in aluminum tank plate into crystal sites hydrogen peroxide physically cannot breach, enabling decade-scale storage without hazardous passivation.



HOW IT WORKS

Annealing 5254 plate at 480 °C drives residual Fe and Cu into Al_3Fe and $Al_6(Fe,Mn)$ intermetallics [26], [32] whose formation enthalpies exceed the oxidation potential H_2O_2 can deliver, locking catalytic metals in lattice sites the oxidizer cannot breach. Current practice controls only composition and surface chemistry [34]; this opens a third axis – microstructural site occupancy – set by a single upstream heat treatment before cold-rolling to flight strength. Intermetallics formed above 400 °C are

"If this was [available to every engineer] they would [use it] daily." "...ideas which no one in my team could come up with." — R&D leader, Fortune 500 automotive

BUILT ALONGSIDE Covestro · Ursa Major · Toyota · Bosch · Conservation X Labs · and a public P&G challenge

WHO TO BRING US

Industrial R&D with a *named, stuck problem* — or an underused capability.

MATERIALS & CHEMICALS

Covestro · PPG · BASF · Dow · DuPont · 3M · Henkel · Solvay · Evonik · SABIC · Arkema

CONSUMER GOODS & FOOD

P&G · PepsiCo · Chobani · Unilever · Nestlé · Mondelez · Mars · L'Oréal · Kraft Heinz

AEROSPACE, AUTOMOTIVE & INDUSTRIAL

Ursa Major · Toyota · Bosch · Mitsubishi · Wabtec · Alcoa · Westinghouse · RTX · GM · Ford

partner · in conversation · plain: target

THE RIGHT PERSON

VP or Head of R&D · CTO, Chief Science or Chief Innovation Officer · open-innovation and alliance-management leads · the BD lead looking for new markets for an existing material or IP.

THE SIGNAL IT'S A FIT

A problem their team has been on for years · a paid-for capability whose best markets they can't see · they've tried the LLMs and been underwhelmed · an open-innovation mandate and appetite for a pilot.

HOW THIS CAN HELP YOU

Walk into the first meeting with three concrete directions – and the CMU people who could pursue them.

01 Start the conversation

A company's interest is nebulous, or they have a capability and no idea where it fits. Send it to us. We run it through the engine and hand you back concrete directions and the faculty across CMU who could pursue each one – a reason to meet, not a brochure.

02 Sponsored research & workshops

Through the lab: a half-day ideation session with their R&D team on a named problem, or a scoped sponsored project on the thing they've been stuck on. Their experts stay in the loop; our students and postdocs do the work.

03 Matchmaking across CMU

Run it in reverse: from a company's capability to the problems across campus it could solve. It makes your outreach strategic – the intro comes with a match, not just to us.

For production deployment inside a company there is also a CMU spinout, Analogical Engines, built on this research – so an inquiry can be routed to the right door: exploratory work through the lab, deployment through the company.

BRING ME A COMPANY AND A PROBLEM

nkittur.github.io/unexpected-places

Case studies, what we offer, who to bring · nkittur@cs.cmu.edu

Happy to talk to any of you.

Aniket Kittur